

GLOBAL RENEWABLE ENERGY REPORT

Trends, Investments & Outlook 2019–2024

Annual Research Edition — June 2024

Global Energy Analytics Division

Executive Summary

The global transition to renewable energy accelerated dramatically between 2019 and 2024, driven by falling technology costs, supportive policy frameworks, and record levels of private and public investment. Total installed renewable capacity surpassed **4,675 GW** by end-2024, representing a **48% increase** over the five-year period. Solar photovoltaic remained the fastest-growing technology, while hydropower and wind continued to underpin baseload and dispatchable supply in key markets.

Clean energy investment reached a record **USD 623 billion** in 2024, with Asia-Pacific accounting for 45% of total global capacity additions. Despite macroeconomic headwinds, including supply-chain disruptions and rising interest rates, the sector demonstrated remarkable resilience. This report synthesises capacity data, financial flows, and regional analysis to provide a comprehensive picture of the current landscape.

4,675 GW	USD 623B	+48%	163 Countries
Total Installed Capacity	2024 Investment	5-Year Growth	Active Markets

1. Installed Capacity by Technology

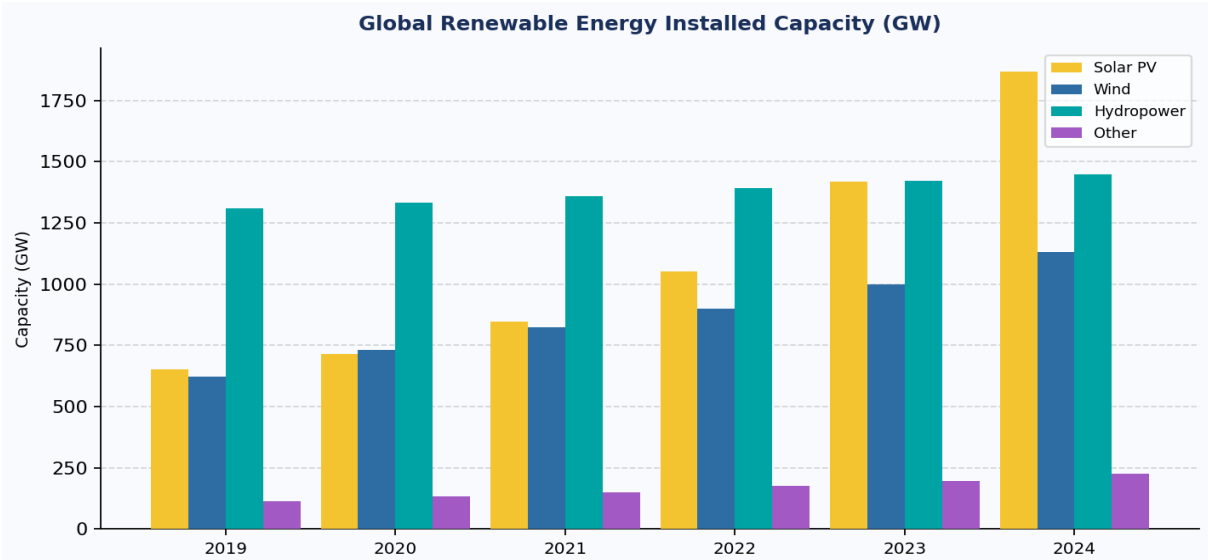


Figure 1 — Grouped bar chart showing installed capacity (GW) for the four main renewable technologies from 2019 to 2024. Solar PV growth outpaced all other sources.

2. Investment Trends

Clean energy financing has grown at a compound annual rate of **17.2%** since 2019, fuelled by declining levelised costs of electricity (LCOE) and an expanding universe of green bonds and sustainability-linked instruments. Solar PV attracted the largest share of new capital in 2024, overtaking wind for the first time in absolute investment terms.

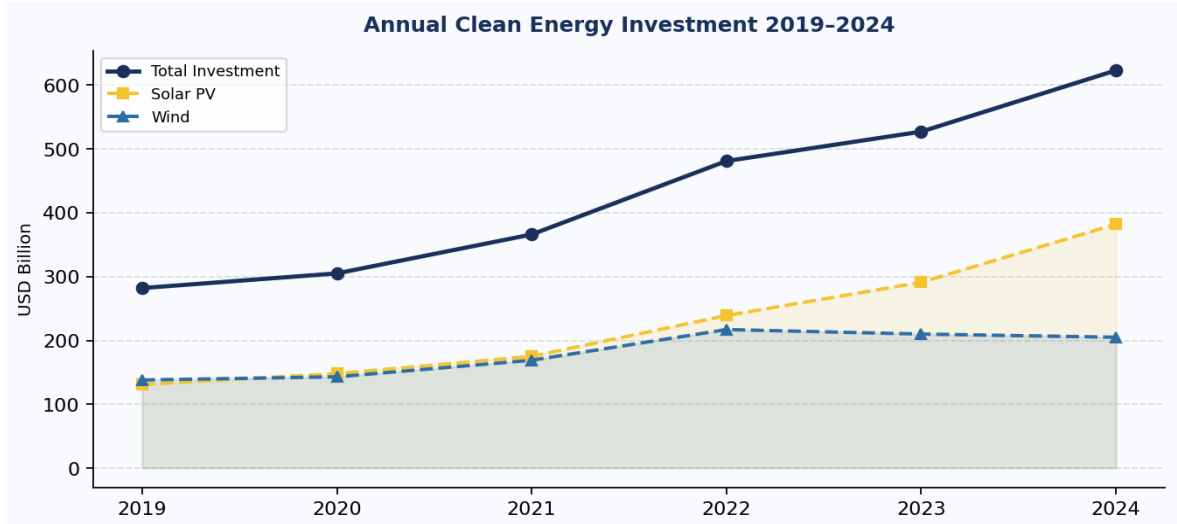


Figure 2 — Total clean energy investment and breakdown between Solar PV and Wind (USD Billion), 2019–2024. Shaded areas highlight each technology's contribution.

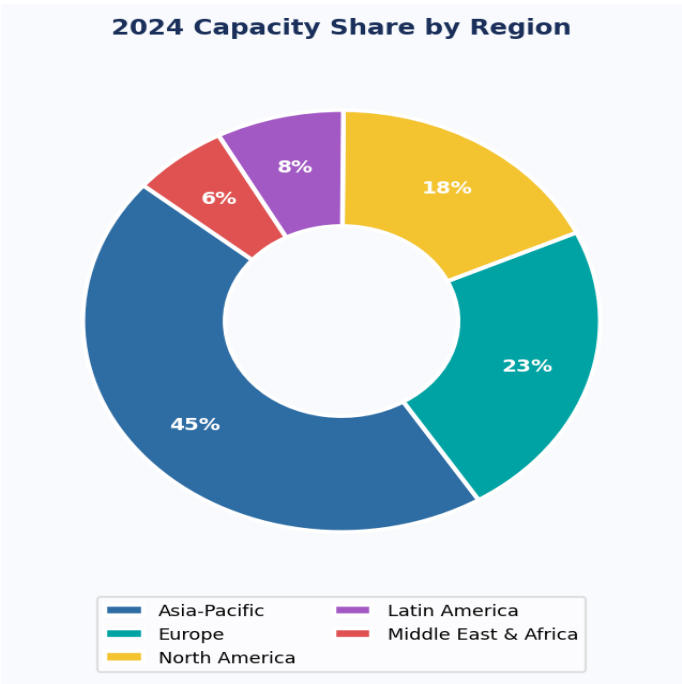
3. Top 10 Markets by Installed Capacity (2024)

Rank	Country	Total (GW)	Solar (GW)	Wind (GW)	Hydro (GW)	YoY Growth
1	China	1,572	590	441	421	+12.4%
2	United States	452	179	148	103	+9.1%
3	India	289	82	44	146	+16.3%
4	Germany	181	81	67	5	+8.7%
5	Brazil	178	41	25	99	+7.2%
6	Japan	152	92	5	49	+6.8%
7	United Kingdom	119	17	28	4	+9.3%
8	Australia	94	38	12	8	+18.1%
9	Spain	87	32	31	20	+11.0%
10	Canada	83	16	15	81	+4.2%

Table 1 — Top 10 countries ranked by total renewable installed capacity (GW) as of December 2024, with technology breakdown and year-on-year growth rate.

4. Regional Capacity Distribution

Asia-Pacific dominates global renewable capacity, driven overwhelmingly by China and India's ambitious deployment programmes. Europe maintains a strong second position underpinned by the EU's REPowerEU framework, while North America benefits from the US Inflation Reduction Act stimulus. Emerging markets in Latin America and the Middle East & Africa are growing rapidly from a lower base.



Technology	2019 LCOE (\$/MWh)	2024 LCOE (\$/MWh)	Change (%)
Solar PV (utility)	\$68	\$36	-47%
Onshore Wind	\$53	\$33	-38%
Offshore Wind	\$115	\$82	-29%
Hydropower	\$47	\$43	-8%
Geothermal	\$74	\$71	-4%

Figure 3 — 2024 capacity share by region (donut chart). Table 2 — Levelised cost of electricity (LCOE) by technology, 2019 vs. 2024.

5. Outlook & Key Recommendations

Scale Battery Storage:	Grid-scale storage must triple by 2030 to absorb rising variable renewable generation and replace firm fossil-fuel capacity. Policy mechanisms should de-risk long-duration storage projects through capacity markets and contracts for difference.
Electrify End-Use Sectors:	Buildings, transport, and industrial heat represent the largest untapped opportunity. Coupling electrification with clean power grids could eliminate 30% of global CO2 emissions within a decade.
Strengthen Grid Infrastructure:	Investment in transmission and distribution infrastructure must keep pace with generation additions. Cross-border interconnectors are essential for regional balancing and energy security.

**Accelerate
Emerging-Market
Finance:**

Blended finance mechanisms and concessional capital from multilateral development banks are critical to unlocking the massive renewable potential in Sub-Saharan Africa and South-East Asia.

Data sources: IRENA Renewable Capacity Statistics 2024; BloombergNEF New Energy Outlook 2024; IEA World Energy Investment 2024. All figures are provisional and subject to revision. This report is for informational purposes only.